Multilevel clustering

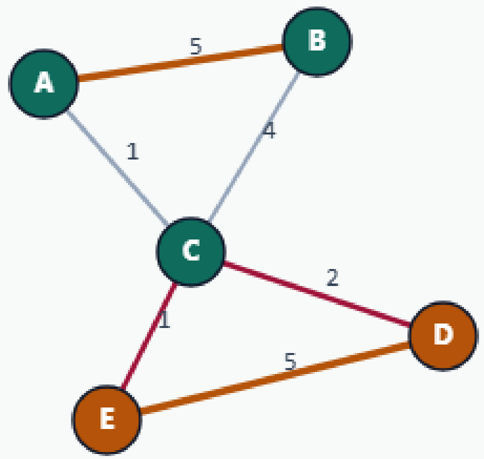
Multilevel clustering coarsens, partitions, then refines

Coarsen with greedy merge

MULTILEVEL CLUSTERING

Coarsen with greedy merge

Step 1 / 5 · coarsen · module03-01-multilevel-clustering



Explanation

Multilevel starts by contracting the graph. Greedy pair merge to coarseK=2 builds a projected bipartition of the original nodes.

- Coarsen → initial partition → refine
- Here: greedy to K=2
- Labels become FM sides 0/1

```
coarseK=2
projected communities ready
```

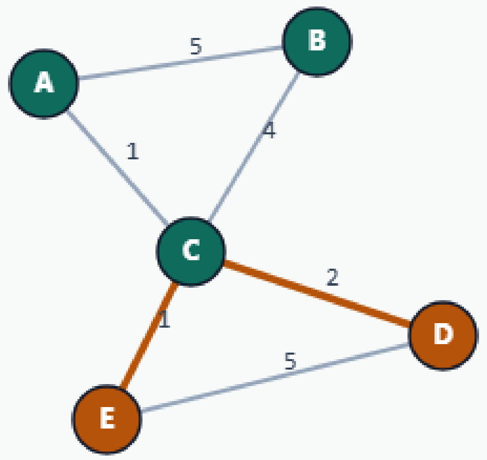
Coarsen with greedy merge

Project to a bipartition

MULTILEVEL CLUSTERING

Project to a bipartition

Step 2 / 5 · project · module03-01-multilevel-clustering



Explanation

Map the two coarse clusters onto sides 0 and 1. This seed is already near-optimal on TINY_GRAPH — unlike BAD_SEED's cut of 12.

- Projection preserves membership
- Contrast: bad seed AE|BCD
- Refinement polishes locally

```
projected cut ≈ 3
BAD_SEED cut = 12
```

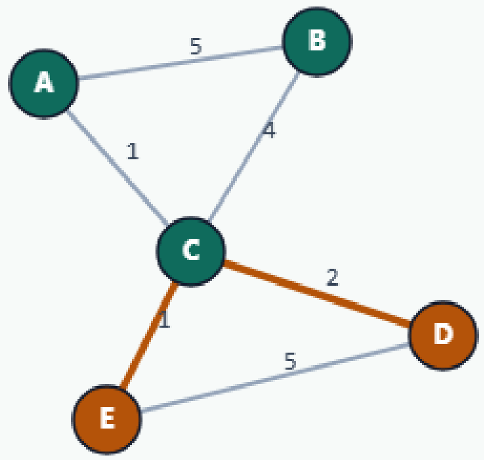
Project to a bipartition

FM-refine on the fine graph

MULTILEVEL CLUSTERING

FM-refine on the fine graph

Step 3 / 5 · fm-refine · module03-01-multilevel-clustering



Explanation

Run Fiduccia–Mattheyses on the projected seed. On this instance refinement keeps ABC vs DE and renames sides to P0/P1.

- Single-vertex moves with rollback
- Output labels: P0 and P1
- cutsize stays 3

```
FM refine
labels → P{side}
```

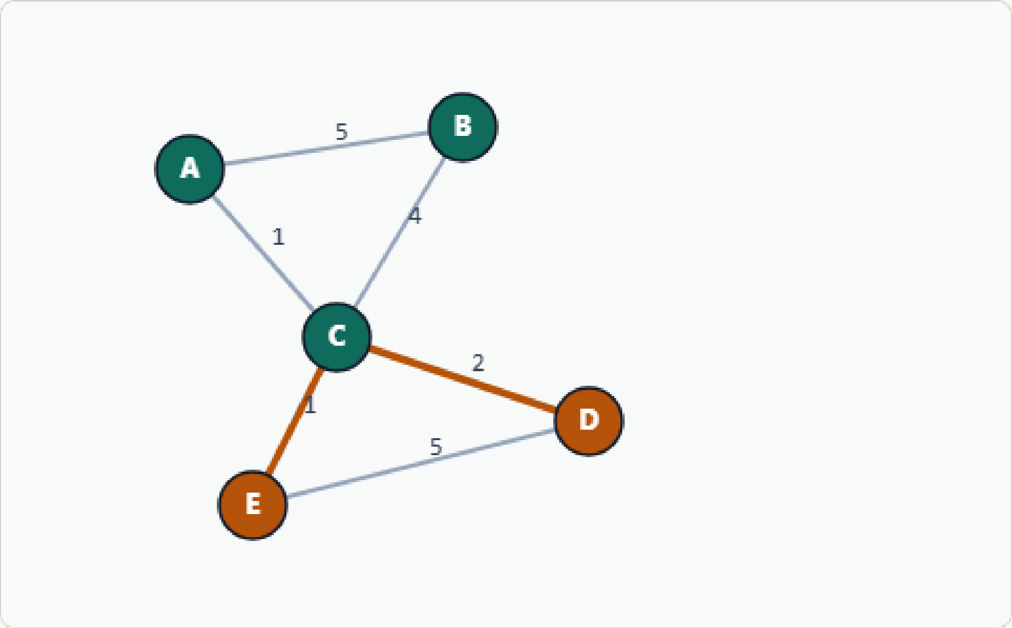
FM-refine on the fine graph

Golden: cutsize 3

MULTILEVEL CLUSTERING

Golden: cutsize 3

Step 4 / 5 · final · module03-01-multilevel-clustering



Explanation

Final communities ABC|DE with P0/P1 labels. Multilevel's value shows more on larger graphs; here it demonstrates the V-cycle shape.

- parts: ABC|DE
- cutsize: 3
- Beats refining a random bad seed alone

```
cutsize: 3
P0=ABC P1=DE
```

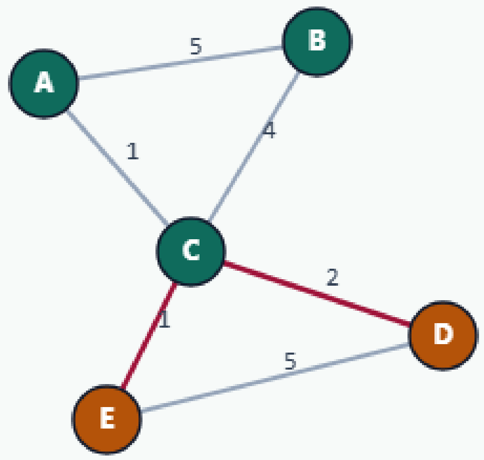
Golden: cutsize 3

Multilevel mindset

MULTILEVEL CLUSTERING

Multilevel mindset

Step 5 / 5 · takeaway · module03-01-multilevel-clustering



Explanation

Coarsen for global structure, refine for local cut. Real placers nest this inside V-cycles with hyperedges and tighter balance — this lab is the skeleton.

- Coarsen = greedy here
- Refine = FM here
- Next labs: hypergraph + EDA objectives

```
Starter golden: cut=3
```

Multilevel mindset

Browser lab track

- In the browser lab, run multilevel and compare with a plain greedy seed
- Clear challenges for the refined ABC|DE result

Implement track

- Run multilevel on the tiny graph and confirm cutsize three
- Re-implement project-and-refine until the unit test passes

Implement track — try these

```
export PYTHONPATH=../common

python ../common/solvers.py examples/tiny_graph.json --mode multilevel
```

Pitfalls to watch

- Skipping refinement leaves coarsening artifacts
- Projecting labels incorrectly scrambles the fine-level seed
- Too aggressive coarsening can erase structure

Your turn

- Match the golden, finish the checklist and quiz, then continue to hypergraph clustering